17. Answer : option(d)

$\left[Fe(en)_3\right]^{2+}\left(PO_4^{3-}\right)$

18. Answer : option(c)

(a) $Zn^{2+}\left(d^{10} \Rightarrow \text{diamagnetic}\right)$

(b) $Co^{3+}\left(d^6 \text{Low spin} \Rightarrow t_{2g}^{\ 6}, e_g^{\ 0}\ ;\ \text{diamagnetic}\right)$

(c) $Ni^{2+}\left(d^8 \text{Low spin} \Rightarrow t_{2g}^{\ 6}, e_g^{\ 2}\ ;\ \text{paramagnetic}\right)$

(d) $\left[Ni(CN)_4\right]^{2-}\left(dsp^2\ ;\ \text{square planar, diamagnetic}\right)$

19. Answer : option(c)

$\left[Co(NH_3)_3(Cl)_3\right]$

20. Answer : option(d)

$V^{2+}\left(t_{2g}^{\ 3}, e_g^{\ 0}\ ;\ \text{CFSE} = 3 \times (-0.4)\Delta_0 = -1.2\Delta_0\right)$

$Ti^{2+}\left(t_{2g}^{\ 2}, e_g^{\ 0}\ ;\ \text{CFSE} = 2 \times (-0.4)\Delta_0 = -0.8\Delta_0\right)$

Statements given in option (a) ,(b), and (c) are wrong.

The current statements are

(a) since, the crystal field stabilization in more octahedral field , octahedral complexes are more stable than square planar complexes.

(b) $\left[FeF_6\right]^{4-}$ – (Fe^{2+} – d^6 High Spin – $t_{2g}^{\ 4}, e_g^{\ 2}$; CFSE = $4 \times (-0.4) + (0.6) \times 2 + P$)

$\left[Fe(CN)_6\right]^{4-}$ – (Fe^{2+} – d^6 Low Spin – $t_{2g}^{\ 6}, e_g^{\ 0}$; CFSE = $6 \times (-0.4) + 3P$)

UNIT-6

1. c) both covalent crystals

2. b) AB_3

number of A ions $= \left(\frac{N_c}{8}\right) = \left(\frac{8}{8}\right) = 1$

number of B ions $= \left(\frac{N_f}{2}\right) = \left(\frac{6}{2}\right) = 3$

simplest formula AB_3

3. b) 1:2

if number of close packed atoms =N; then,

The number of Tetrahedral holes formed = 2N

number of Octahedral holes formed = N

therefore N:2N = 1:2

Answers.indd 240 3/29/2019 6:43:49 PM